BITS College
Department of Software Engineering
SWEN466: Knowledge Discovery and Data Mining

Assignment 1: Data Quality Analysis and ERD Design

Deadline: 26 October 2026, 9:00 AM

Objective

Explore the Olist dataset using DuckDB by completing the five SQL exercises in the **01_explore_source_data** class notebook and constructing an entity-relationship diagram (ERD) for five Olist tables using crow's foot notation. Identify and validate primary keys, foreign keys, and relationship cardinalities through SQL queries rather than inferring them from column names alone.

Data

You have five CSV files, loaded into the **raw** schema: **customers**, **orders**, **order_items**, **products** and **sellers**.

Tasks

1. **Explore.**
    a. Run **DESCRIBE** and **SELECT * ... LIMIT 5** on each table. Write down what one row represents (the grain).
    b. **Primary keys.** For each table, the number of rows must equal the number of distinct values, with no nulls. One table needs a key made of two columns.
    c. **Foreign keys.** For each relationship, the parent table, the child table and the column that links them. This is done with an orphan check using **ANTI JOIN**.
2. **Measure the cardinality.** For each relationship, find the minimum and maximum number of child rows per parent row. Use this to decide between "exactly one", "zero or one", "one or many" and "zero or many".
3. **Draw the ERD.** Show every table with its columns, mark PK and FK, and connect the tables with crow's foot symbols and a verb on each line (for example "places").
4. **Write a short note** (3 to 5 sentences) answering the following questions:
    a. Two columns in the customers table look like customer identifiers. How do they differ, and which one would you use to count customers?
    b. Which table is a junction table, and which relationships does it resolve?

c. Name one relationship where the data is stricter than your diagram, and one where the circle on your diagram is justified by the data

Tools

Use any of these: draw.io, Mermaid, or pen and paper with a photo.

Hints

- A foreign key has the same name as the primary key it points to, but check the data: matching names are a clue, not proof.
- A table can be a parent in one relationship and a child in another. Decide roles one relationship at a time.
- A line with a circle means "may have none". Look for parents with no children in the data before you draw the circle.
- Label the primary key of a composite key on every column that is part of it.

Questions to ask while you work

Each table	Each relationship	Diagram
What does one row represent?	Which table holds the foreign key? That one is the child.	Is any table linked to several others? What role does it play?
Which column or columns identify it uniquely, and how do I prove it?	Can a child row exist without a parent? Can a parent exist without children?	Is there a many-to-many relationship hidden between tables? Which table solves it?
Are there nulls in the key?	What is the largest number of children one parent has in the data?	Does my diagram describe what the data allows, or what the data happened to contain?
	Are there child rows whose parents are missing (orphans)?	

Deliverables – What to submit

- The ERD as an image or PDF.
- The notebook with SQL queries for the data quality checks. Replace **SELECT 'write your query here'** in Exercises 1–5 with one DuckDB query each, reading only from **raw.<table>**.
- Your short note (it can be a markdown file in the repo).

Commit these files to your **kddm_lab** repository and submit the link of your repo on the e-learning platform.